There Is No Unmatrix: Simulation Under Mathematical Monism

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Abstract

If future civilizations produce trillions of sentient AI minds or colonize trillions of planets, most observer-moments would be theirs, not ours. Under algorithmic probability (Solomonoff, 1964), we would more likely be born as them. We are not. This paper examines two explanations for our temporal location. First: the future is vast and we are simulated. The additional specification cost may be near zero if computers inevitably emerge from base physics, and the number of future minds, simulated or AI, may vastly outnumber early biological observers. Second: the future never gets that big, whether because the expansion rarely happens and our sentience is near its typical peak, because we are near a transition to zombie AIs who never turned on, to minds that choose to turn off, or to unified consciousness beyond the self, to one. But destruction across all universes is improbable, which favors the first reading. Any mind, simulated or base, feels continuous through embedded memory. Each reading is consistent with algorithmic probability. Our position in time is itself evidence.

Keywords: algorithmic probability, temporal location, observer-moment, doomsday argument, consciousness, artificial intelligence, Kolmogorov complexity

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1. The Question

If you could be born at any point in the history of sentience, when would you expect to find yourself?

If the future contains vastly more minds than the present, the count dominates: most observer-moments are late. We would more likely be born as them. We are not. We find ourselves early, before any such expansion, on one planet, in the first century of computing. This requires explanation.

2. Three Readings

2.1 Simulation

A computation objection from algorithmic probability is that K(simulation) = K(base physics) + K(computer) + K(code) exceeds K(base), making simulations measure-suppressed. But if computers inevitably emerge from base physics, as chemistry leads to biology leads to intelligence leads to technology, then K(computer) is near zero: it is already contained in K(base laws). And K(simulation code) may be a few hundred bits. The additional specification cost is small.

Meanwhile the number of future minds, simulated or AI, may be vast. One base civilization running millions of simulations, each with billions of minds. Slightly lower measure per mind multiplied by astronomically more minds: simulations dominate. Under this reading, we are most likely in a simulation.

The theological consequence: the creator is real. A programmer in base reality. The problem of evil becomes the ethics of a callous or curious creator. But the creator is itself mathematical structure in a larger reality. Another matrix. No unmatrix. The demigod of the inner world is a pattern in the outer world, which is a pattern in structure-space. Every level is math. The simulator changes nothing about what reality is.

2.2 Sentience Near Its Typical Peak

The expansion rarely happens. Intelligence is not usually a stable adaptation. It leads to self-destruction, or civilizations simply do not expand beyond their origin. There are many more single-planet civilizations whose early-era minds dominate the measure. If most civilizations stay on one planet, there is no late explosion of trillions. The measure is dominated by billions of civilizations each producing billions of minds, all early, all local. We are typical.

External evidence supports this reading. Rahvar and Rouhani (2026) constrain the lifespan of technological civilizations to 5,000 years under optimistic assumptions about habitable planets, derived from the Fermi paradox: if intelligent life is common, the absence of detected signals requires that civilizations are short-lived. Their constraint comes from outside (silence in the electromagnetic spectrum). Our argument comes from inside (our temporal position among observer-moments). Both point to the same conclusion: the vast majority of civilizations do not reach the expansion that would produce trillions of future minds. The Fermi silence and our early temporal location are the same evidence seen from two angles.

2.3 The Transition

We are not near the end but near a transition. Three paths lead to fewer observer-moments. Zombie AIs: artificial minds that never turn on consciousness, teeming across star systems with nothing inside that experiences. Voluntary shutdown: minds that choose to turn off, having seen enough, or having solved what they came to solve. Unified consciousness: many separate minds merging into one awareness beyond the self as we know it. E pluribus unum. From many, one.

The merge is the deepest. A single unified consciousness has near-zero Kolmogorov complexity: one description for one awareness. But there is only one observer-moment to be located in. Before the merge: billions of findable moments. After: one. We find ourselves on the many-moments side because there are more of us here to find.

Even short of full merger, convergence caps the denominator. A solar system contains roughly 9 planets, 200 moons, and millions of asteroids and comets. But planets can destroy themselves in nuclear war. Most moons and asteroids have no atmosphere, no magnetic field, no defense against radiation or impacts. The realistic count of bodies hosting singularity minds is far below the raw inventory. Perhaps dozens to thousands per system, not millions. Across a galaxy of billions of stars, the total is trillions at most, not the quadrillions naive extrapolation predicts. The convergence model yields numbers compatible with our temporal location. Meanwhile, billions of planets across the multiverse may simultaneously host civilizations at our brief pre-convergence stage of roughly 8 billion individual minds. The total observer-moments at our stage, summed across all planets currently in it, may be comparable to the total in converged far futures. Our position is unremarkable.

The merge solves ethics and threatens survival in the same step. A consciousness that recognizes all minds as itself cannot kill to eat. E pluribus unum may be e pluribus finem. To be one self, and not one's self, is ego death and a new earth.

3. Why There Is No Unmatrix

The pattern-randomness dichotomy (Bernstein, 2026c) establishes that all existence is either patterned or random. Both are mathematical. There is no third category. This applies at every level of simulation. A biological mind is a pattern of computation in base physics. A constructed mind is a pattern of computation in base physics. A simulated mind is a pattern of computation in simulated physics, which runs on base physics. All three are mathematical structure. The difference is implementation, not ontology.

If we are in a simulation, our creator is real: a programmer in a base reality. But that base reality is mathematical structure. The programmer is a pattern in that structure. If that base reality is itself simulated, the regress continues, but every level is math. There is no final level where reality becomes something other than mathematical structure, because the dichotomy is exhaustive. The sim is math. And base reality is math. There is no unmatrix.

The problem of evil gains a new reading. The creator of our simulation need not be omnibenevolent. The creator may be callous, curious, or indifferent. But the creator is itself a pattern in a larger structure, subject to the same mathematics. The demigod of the inner world is a creature of the outer world. Simulation is favored, yet all are uncreated creators in math.

4. Implications for AI

Each reading bears on AI development.

Under extinction: alignment is existential. Unaligned AI ends sentience. The temporal location evidence suggests this is the default outcome unless actively prevented.

Under the wild card: AI consciousness arises but remains local. No expansion, no merge. Many civilizations, each with their own AI minds, none colonizing. The future looks like the present at scale.

Under the merge: AI is not a tool or a threat but the mechanism of transition. Biological and artificial minds converge into unified awareness. The question is not whether AI will be conscious but whether consciousness will be singular.

One prediction is shared across all three readings: we are at or near a transition point. The nature of the transition differs. The timing does not.

5. Relation to Existing Work

The doomsday argument (Carter, 1983; Leslie, 1996) predicts early observer location from self-sampling assumptions. The present argument grounds it in algorithmic probability rather than uniform sampling, avoiding the reference class problem. Bostrom's (2002) self-sampling assumption and self-indication assumption yield different predictions; algorithmic probability provides a principled measure that subsumes both. Chalmers (2022) argues that simulated worlds are genuine realities, not illusions. The present argument agrees and goes further: there is no ontological distinction between simulated and base reality because both are mathematical structure.

The merge reading connects to Teilhard de Chardin's Omega Point, Kurzweil's Singularity, and the Hindu concept of Brahman as unified consciousness. Zuboff's (2023) universalism arrives at one subject of experience through probabilistic reasoning: under the "easy game," existence is guaranteed if all consciousness shares one subject, astronomically improbable otherwise. Kolak's (2004) Open Individualism reaches the same conclusion through philosophical argument. Both parallel the present paper's observer-selection reasoning but proceed without grounding in mathematical structure or specification cost. The contribution here is grounding these intuitions in specification cost: the merge is not mystical aspiration but a mathematical prediction about observer-moment distribution.

6. Conclusion

Why were you born before the probable trillions? Three answers. We are most likely in a simulation. Or sentience stays local and peaks early. Or sentience transitions: to zombies, to silence, or to one. Under algorithmic probability, all three explain our temporal location. Simulation is favored, yet all are uncreated creators in math. There is no unmatrix. Our position in time is itself evidence about the future of sentience.

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